ANSWER KEY & MARKING SCHEME SET 1

Chapter 1: **Exploring Substances: Acidic, Basic and Neutral** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** acidic	Sour taste is a property of acids.
2	**(a)** sulphuric acid	Sulphuric acid is a strong mineral acid.
3	**(b)** base	Bases turn red litmus blue.
4	**(a)** basic	Bases are bitter and slippery.
5	**(a)** H_2SO_4	It contains two hydrogen atoms.
6	**(c)** do not change the colour of litmus	They are neither acidic nor basic.
7	**(d)** lemon juice	Lemon juice contains citric acid.
8	**(d)** $NaOH$	A common strong base.
9	**(b)** base	Turmeric becomes red with soap solution.
10	**(c)** baking soda solution	Baking soda is a base.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Substances that show the nature of a solution by colour change (1); Litmus and turmeric (1).	1+1
12	Baking soda (1); Lime water or soap (1).	1+1
13	HCl (1); $NaOH$ (1).	1+1
14	Acids turn blue litmus red (1); Bases turn red litmus blue (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Dip blue and red litmus (1); Red from blue means acid; blue from red means base (1); No change means neutral (1).	1+1+1
16	Some are corrosive (1); Some are poisonous (1); Use indicators instead (1).	1+1+1
17	Bacteria make acid from sugar (1); Acid damages enamel (1); Toothpaste is basic (1).	1+1+1
	OR: Litmus: red in acid, blue in base (1); Turmeric: yellow in acid, red in base (1); China rose: dark pink in acid, green in base (1).	1+1+1
18	Boil cabbage leaves and filter (1); Add to acid: turns red or pink (1); Add to base: turns green or blue (1).	1+1+1
	OR: It removes sugar that bacteria turn into acid (1); It reduces the acid attack on the enamel (1); It helps to prevent tooth decay (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Collect lemon juice, vinegar, soap, baking soda, salt (1); Test with litmus papers (1); Test with turmeric (1); Test with China rose (1); Record in a table and classify (1).	1+1+1+1+1
	OR: Citric: lemon (1); Acetic: vinegar (1); Lactic: curd (1); Tartaric: tamarind (1); Ascorbic: amla (1).	1+1+1+1+1
20	Acid + base reaction (1); $HCl + NaOH \longrightarrow NaCl + H_2O$ (1); Salt and water formed (1); Treating acidity (1); Treating acidic soil (1).	1+1+1+1+1
	OR: Mix the soil with water and filter (1); Test the filtrate with litmus papers (1); Blue litmus turns red: acidic soil (1); Red litmus turns blue: basic soil (1); No change: neutral soil (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that acids turn red litmus blue; remember **acid: blue litmus becomes red**, and **base: red litmus becomes blue**.
- Expecting turmeric to change colour with an acid; turmeric changes (to red) only with a base.
- Forgetting that a neutral substance (sugar solution, water) gives *no change* with every indicator.
- Writing the products of neutralisation as only 'salt'; the products are **salt and water** (with heat).
- Mixing up the colours of China rose: acid gives *dark pink*, base gives *green*.
- Believing that every sour thing is dangerous; many edible substances (lemon, curd, vinegar) are weak acids, but laboratory acids must never be tasted.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 1: **Exploring Substances: Acidic, Basic and Neutral** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** $Ca(OH)_2$	It is a base.
2	**(b)** acid	Acidic solutions turn it pink.
3	**(a)** sugar solution	Sugar solution has no acidic or basic character.
4	**(d)** antacid	It contains magnesium hydroxide.
5	**(c)** neutralisation	Acid + base gives salt and water.
6	**(a)** lemon and orange	Citrus fruits contain it.
7	**(d)** $NaOH$	It has OH.
8	**(a)** salt and water	Salt and water.
9	**(a)** vinegar	Vinegar is dilute acetic acid.
10	**(d)** HNO_3	Nitric acid.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Ant injects formic acid (1); Baking soda is a base and neutralises it (1).	1+1
12	Red (1); Soap is a base (1).	1+1
13	Reaction of an acid with a base (1); Acid + base gives salt + water (1).	1+1
14	Stomach has excess acid (1); Antacid is a base that neutralises it (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Hydrochloric acid + sodium hydroxide gives sodium chloride + water (1); Sulphuric acid + sodium hydroxide gives sodium sulphate + water (1); Acid + base gives salt + water (1).	1+1+1
16	Treating acidity (1); Treating ant stings (1); Treating soil (1).	1+1+1
17	The solution is red at first (1); It turns purple when neutral (1); It turns blue with excess base (1).	1+1+1
	OR: Lemon juice: acid (1); Soap solution: base (1); Salt solution: neutral (1).	1+1+1
18	HCl, H_2SO_4, HNO_3 (1); $NaOH$, KOH (1); $Ca(OH)_2$ (1).	1+1+1
	OR: Never taste them (1); Wear gloves and goggles (1); Wash spills with plenty of water (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Acids sour, bases bitter (1); Acids turn blue litmus red (1); Bases turn red litmus blue (1); Bases feel soapy (1); Bases turn turmeric red (1).	1+1+1+1+1
	OR: Some crops need acidic soil (1); Some need basic soil (1); Test soil (1); Treat with lime or compost (1); Better yield (1).	1+1+1+1+1
20	Ant sting: baking soda (1); Indigestion: antacid (1); Soil: lime (1); Base cancels acid (1); Relief or good crops (1).	1+1+1+1+1
	OR: Take dilute HCl (1); Add litmus: red (1); Add NaOH drop by drop (1); Colour changes to purple (1); More base gives blue (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that acids turn red litmus blue; remember **acid: blue litmus becomes red**, and **base: red litmus becomes blue**.
- Expecting turmeric to change colour with an acid; turmeric changes (to red) only with a base.
- Forgetting that a neutral substance (sugar solution, water) gives *no change* with every indicator.
- Writing the products of neutralisation as only 'salt'; the products are **salt and water** (with heat).
- Mixing up the colours of China rose: acid gives *dark pink*, base gives *green*.
- Believing that every sour thing is dangerous; many edible substances (lemon, curd, vinegar) are weak acids, but laboratory acids must never be tasted.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 1: **Exploring Substances: Acidic, Basic and Neutral** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** neutralisation	Acid-base reaction.
2	**(a)** curd	Curd is sour due to lactic acid.
3	**(b)** indicators	They show the nature of the solution.
4	**(c)** sodium chloride	Common salt.
5	**(b)** formic acid	Ants inject formic acid.
6	**(b)** lichens	It is a natural dye.
7	**(b)** vinegar	Wasp sting is alkaline.
8	**(d)** HCl	HCl is hydrochloric acid.
9	**(c)** acid	Acids turn blue litmus red.
10	**(c)** baking soda	A base neutralises formic acid.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Water (1); Sugar solution (1).	1+1
12	Citric acid in lemon (1); Lactic acid in curd (1).	1+1
13	Adding quick lime or slaked lime (1); Base neutralises the acid (1).	1+1
14	It may be acidic (1); Acids harm aquatic life (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Excess hydrochloric acid in the stomach (1); Magnesium hydroxide neutralises it (1); Relieves pain (1).	1+1+1
16	A neutral solution is neither acidic nor basic (1); Examples: water, sugar solution (1); Litmus papers show no change (1).	1+1+1
17	Stomach acid helps digestion (1); Bases like soap clean grease (1); Vinegar removes stains (1).	1+1+1
	OR: Curd, lemon: sour (1); Soap, baking soda: bitter (1); Soapy feel for bases (1).	1+1+1
18	Concentrated acid has more acid in less water (1); Dilute acid has more water (1); Vinegar is dilute acetic acid (1).	1+1+1
	OR: Water (1); Sugar solution (1); Common salt solution (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Litmus (1); Turmeric paste on paper (1); Turns red with base (1); China rose pink in acid (1); Red cabbage changes colours (1).	1+1+1+1+1
	OR: Soap (1); Antacids (1); Cleaning fluids (1); Whitewash (1); Fertilisers (1).	1+1+1+1+1
20	Corrosive burns (1); Wear gloves and goggles (1); Add acid to water slowly (1); Do not taste (1); Wash spills with plenty of water (1).	1+1+1+1+1
	OR: Neutral gives no change (1); Both litmus papers stay the same (1); Turmeric stays yellow (1); China rose stays (1); Water, sugar are examples (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that acids turn red litmus blue; remember **acid: blue litmus becomes red**, and **base: red litmus becomes blue**.
- Expecting turmeric to change colour with an acid; turmeric changes (to red) only with a base.
- Forgetting that a neutral substance (sugar solution, water) gives *no change* with every indicator.
- Writing the products of neutralisation as only 'salt'; the products are **salt and water** (with heat).
- Mixing up the colours of China rose: acid gives *dark pink*, base gives *green*.
- Believing that every sour thing is dangerous; many edible substances (lemon, curd, vinegar) are weak acids, but laboratory acids must never be tasted.

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